

Wellington Region Urban Growth Projection Model

CA-Markov + XGBoost Land Cover Change Model, 2012–2035

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Executive Summary

This project simulates urban growth in the Greater Wellington Region, New Zealand, combining a Markov transition matrix with an XGBoost transition-potential model to project land cover from 2023 to 2035. The model was calibrated on verified historical land cover data (2012, 2018, 2023) from New Zealand's Land Cover Database (LCDB) and validated by simulating a known interval (2018→2023) and comparing the result against the real outcome before trusting it to forecast forward.

The model correctly predicted the scale of growth (7,603 predicted vs. 7,273 actual new built-up pixels, within 4.5%) and identified genuine growth hotspots at a neighbourhood scale roughly 39x better than random chance. Distance to the road network was overwhelmingly the dominant predictor of where growth occurs (81.6% feature importance), well ahead of distance to existing development, slope, or distance to water. Projected forward, built-up area in the region is expected to grow from 1.14% to 1.21% of the AOI by 2035, adding approximately 15.4 km² of new development consistent with the 13.4 km² added over the previous 11 real years (2012–2023).

Note on scope: this is a calibrated, one-shot projection pipeline, calibrate on real data, validate against a held-out real interval, then project once to 2035 rather than an interactive or re-runnable simulator with adjustable scenarios or parameters. The name reflects that scope.

1. Background and Methodology Pivot

This project originally targeted Hyderabad, India, using Landsat 8 imagery classified with spectral index thresholds (NDVI, NDBI, MNDWI) in the absence of hand-labelled ground truth. That approach produced land cover maps that were internally inconsistent across years built-up area appeared to swing implausibly (dropping by nearly half between 2013 and 2019, then partially recovering by 2024) because bare or fallow agricultural land and impervious urban surfaces are spectrally similar in a semi-arid, dry-season context, and auto-labelled training data had no mechanism to correct for it. Several fixes were attempted pooling training samples across years into a single classifier, and adding a texture-based discriminator to separate heterogeneous built-up fabric from uniform bare fields, and each improved the picture without fully resolving the underlying problem.

The project was rebuilt on Wellington, New Zealand specifically because the Land Cover Database (LCDB), maintained by Manaaki Whenua and LINZ, provides real, independently verified land cover classifications at five-yearly intervals (1996, 2001, 2008, 2012, 2018, 2023). This removes the ground-truth problem entirely: every land cover class used in this analysis is a verified government classification, not an inferred one. This document reports only the Wellington results; the Hyderabad attempt is not included as it did not reach a trustworthy state.

2. Study Area

The analysis area is the Greater Wellington Region, bounded approximately by 174.38°E–176.31°E and 41.61°S–40.39°S. This is a large region (~8,000 km²) dominated by rural and hill-country land built-up area is only around 1.1–1.2% of the total AOI with the urban conurbation (Wellington City, Lower Hutt, Upper Hutt, Porirua) and the separate Kapiti Coast settlement (Paraparaumu) concentrated along the harbour and valley corridors. All analysis was performed at 30m resolution on a grid of 4,503 × 5,499 pixels, projected to EPSG:2193 (NZTM2000).

3. Data Sources

Dataset	Source	Use
Land Cover Database (LCDB v6.0)	LINZ / Manaaki Whenua, via LRIS Portal	Land cover classification, 2012/2018/2023 editions
NZ Road Centrelines (Topo, 1:50k)	LINZ Data Service	Distance-to-roads driver
NZ Place Names (NZGB)	LINZ Data Service	City/town labels for final map
SRTM 30m DEM	USGS, via Google Earth Engine	Slope driver

4. Methodology

4.1 Land Cover Classification

LCDB's approximately 33 detailed land cover classes were reclassified into four categories for the 2012, 2018, and 2023 editions:

- Water — lakes, rivers, estuarine open water
- Vegetation — indigenous and exotic forest, scrub, wetland vegetation, tussock, bare natural surfaces (rock, sand, snow)
- Pasture/Agriculture — cropland, orchards/vineyards, high and low producing grassland
- Built-up — settlement, urban parkland, transport infrastructure, surface mines/dumps

Each edition was rasterized to the common 30m grid described above.

4.2 Driver Variables

Four physical driver variables were computed for the model: distance to existing built-up area, distance to water, slope (derived from the SRTM 30m DEM), and distance to the road network (from LINZ's official Topo50 road centrelines). Distance-to-roads was initially omitted, mirroring a similar decision made for the Hyderabad attempt where OpenStreetMap's Overpass API repeatedly failed; New Zealand's LINZ Data Service provided a reliable authoritative alternative, so roads were added back in and turned out to dominate the model (see Section 4.4).

4.3 Markov Transition Matrix

A transition matrix was computed by cross-tabulating land cover class at one date against the next, first 2012 against 2018 for initial calibration, then 2018 against 2023 for the final model used in the forward projection, since the more recent interval better reflects current growth dynamics. Each row of the matrix gives the probability of a pixel in a given class transitioning to each other class over that interval.

4.4 XGBoost Transition-Potential Model

An XGBoost binary classifier predicts, for every pixel not already built-up, its probability of converting to built-up over the calibration interval, using the four driver variables above as features. In the calibration run, distance-to-roads accounted for 81.6% of feature importance, versus 14.0% for distance-to-built-up, 2.5% for distance-to-water, and 2.0% for slope confirming that proximity to transport infrastructure is by far the strongest determinant of where new development occurs in this region, consistent with the wider urban growth modelling literature.

4.5 Forward Simulation to 2035

The model was recalibrated on the most recent complete real interval (2018→2023) and projected forward in three approximately 4-year steps (2023→2027→2031→2035). At each step, distance-to-built-up and distance-to-water were recomputed from the evolving simulated state, not the static 2023 baseline, so that new growth influences where the next round of growth is likely to occur, consistent with how cities actually expand outward from existing

development rather than in isolated patches. Water was excluded from the convertible land pool, since land reclamation is a real but separate process that these generic distance-based drivers have no basis for predicting. The number of new built-up pixels allocated at each step was clamped to the remaining available (non-built, non-water) land, in case future demand were to exceed it.

5. Validation

Before projecting forward, the calibrated model was used to simulate the known 2018→2023 interval, and the result was compared against the real 2023 LCDB edition.

Metric	Value	Interpretation
Expected new built-up pixels (Markov)	7,603	Aggregate demand implied by the transition matrix
Actual new built-up pixels (real 2023 LCDB)	7,273	Within 4.5% of the predicted quantity
Held-out ROC-AUC	0.9978	Strong ranking quality on unseen data
Pixel-exact location hit rate	5.1%	Modest — exact-pixel prediction is an unreasonably strict test
300m neighbourhood hit rate	25.4%	~39x better than the ~0.6% expected by random chance
Cohen's Kappa (built vs non-built, whole map)	0.9744	Strong whole-map agreement

The aggregate growth quantity prediction is strong. Exact-pixel spatial precision is more modest, which is expected: at 30m resolution, predicting the precise cell a specific subdivision or council consent will land on is an unreasonably strict test, since actual development location is also shaped by zoning and planning decisions that no generic distance-based surface can fully capture. The standard way this literature assesses spatial performance is at a coarser neighbourhood scale (Pontius et al.), which is why the 300m-cell result 39x better than chance, is the more meaningful spatial validation figure here.

6. Results: 2035 Projection

Built-up area is projected to grow from 1.138% of the AOI in 2023 to 1.207% by 2035, adding approximately 15.4 km² of new development across the three simulated steps (+5,693, +5,690, and +5,686 pixels respectively, a notably steady, non-explosive rate). This is proportionally consistent with the 13.4 km² of real, verified growth observed over the previous 11 years (2012–2023). The map below shows the historical (2012 baseline, real 2012→2023 growth) and projected (simulated 2023→2035 growth) pattern; growth traces closely along the road corridors connecting Wellington City, Lower Hutt, Upper Hutt, Porirua, and Paraparaumu, consistent with the model's heavy reliance on the road's driver.

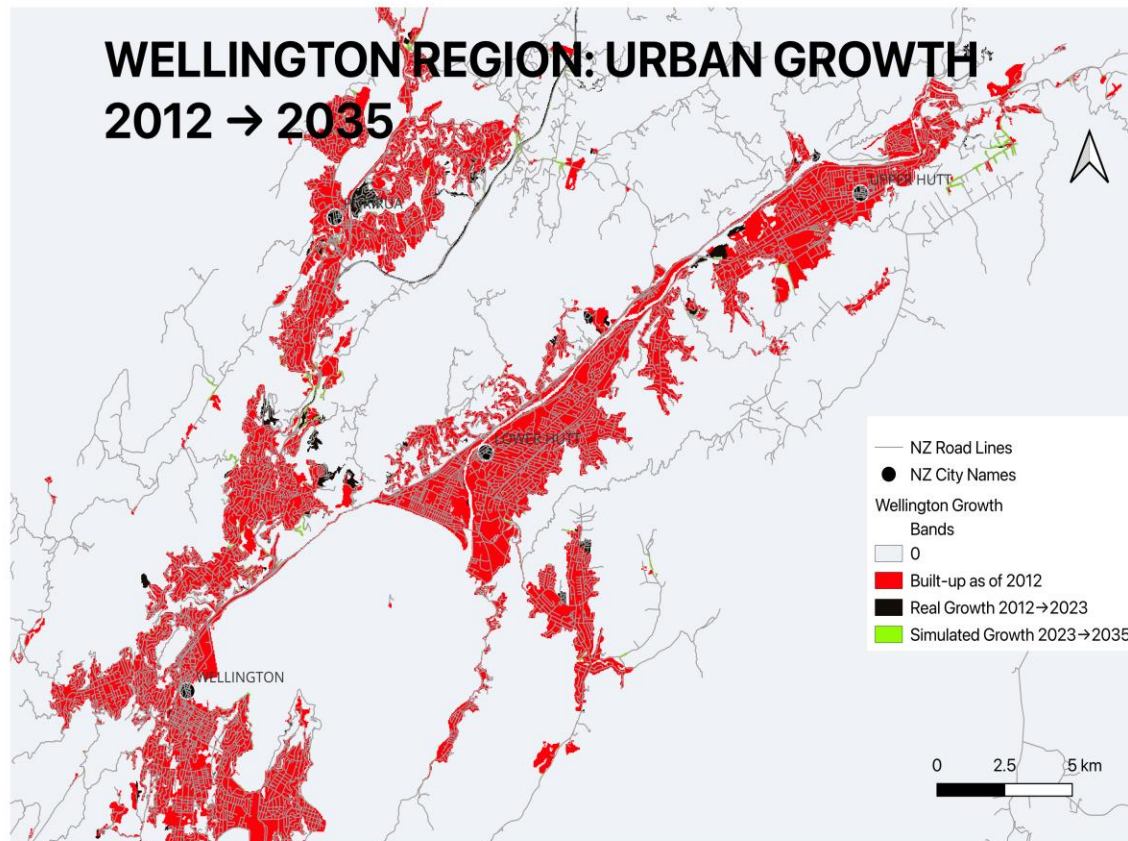


Figure 1. Wellington Region urban growth, 2012 baseline through simulated 2035 projection. Black = real growth 2012→2023; green = simulated growth 2023→2035. Data: LINZ LCDB v6.0, LINZ Topo50, LINZ NZGB place names.

7. Limitations

- The AOI covers the full Greater Wellington Region, which is dominated by rural land; built-up area is only ~1.1–1.2% of the total. Display maps are cropped to the urban extent for legibility, but all analysis covers the full region.
- The roads dataset is a current snapshot applied uniformly to the 2012, 2018, and 2023 states, since no historical roads time series was available. Road networks change far more slowly than individual buildings, making this a reasonable approximation, but not an exact one.
- Pixel-exact spatial precision (5.1%) is modest despite strong aggregate accuracy. This reflects both the inherent difficulty of the task and the fact that specific development sites depend on zoning and consent decisions not captured by physical distance drivers alone.
- The forward projection assumes the calibrated 2018–2023 transition rates remain stationary through 2035. Policy changes, zoning reform, or major infrastructure investment could shift future rates in ways the model cannot anticipate.
- Time-step scaling between the 5-year calibration base and the ~4-year projection steps are linear, a standard CA-Markov simplification.

8. Conclusion

Switching the study area from Hyderabad to Wellington traded a data-quality problem that could not be fully resolved with available methods for one with genuine, verified ground truth. The resulting model captures Wellington's real growth dynamics credibly at the aggregate level and meaningfully above chance at the spatial level, with a validated methodology and an honest account of its limitations. The 2035 projection should be read as a plausible, evidence-based scenario consistent with the region's recent growth trajectory, not a precise prediction of specific development sites.